INCH-POUND

MIL-DTL-32005(OS)
23 October 1997

DETAIL SPECIFICATION

CARTRIDGE, IMPULSE, CCU-143/A

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document specifies the requirements for the manufacture, assembly, and packaging of the CCU-143/A impulse cartridges and the methods of inspection and tests upon which product acceptance will be based.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Team (Code 8410T), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

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SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-L-757	Lead Styphnate, Normal
MIL-P-20444	Primer, Percussion, M42 Parts for and Loading, Assembling and Packing
MIL-L-46225	Lead Azide RD-1333 (for Use in Ammunition)

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-1168	Ammunition Lot Numbering
DOD-STD-2101	Classification Of Characteristics

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

PURCHASE DESCRIPTIONS

NAVAL SEA SYSTEMS COMMAND (CAGE Code 10001)

WS 5003	Hexanitrostilbene (HNS)

(Application for copies should be addressed to the Commander, Indian Head Division, Naval Surface Warfare Center (Code 8410P), 101 Strauss Avenue, Indian Head, MD 20640-5035.)

DRAWINGS

NAVAL AIR SYSTEMS COMMAND (CAGE Code 30003)

DL850AS161	Cartridge, Impulse, CCU-143/A
DL841AS287	Detonator Tester Assembly
850AS161	Cartridge, Impulse, CCU-143/A

(Application for copies should be addressed to the Commander, Indian Head Division, Naval Surface Warfare Center (Code 8410P), 101 Strauss Avenue, Indian Head, MD 20640-5035.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR QUALITY CONTROL

ASQC Z1.4	Sampling Procedures and Tables for Inspection by Attributes

(Application for copies should be addressed to the American Society for Quality Control, PO Box 3005, 611 E. Wisconsin Ave., Milwaukee, WI 53201-4606.)

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI Z1.1	Guide for Quality Control Charts
ANSI Z1.2	Control Chart Method of Analyzing Data
ANSI Z1.3	Control Chart Method of Controlling Quality During Production

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM E545	Standard Method for Determining Image Quality in Direct Thermal Neutron Radiographic Examination
ASTM E1742	Standard Practice for Radiographic Examination

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Dr., West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The CCU-143/A impulse cartridges shall be manufactured in accordance with Drawing DL850AS161 and all documents and drawings listed thereon. Drawings listed in this specification are included as part of the data package defined by Drawing DL850AS161.

3.2 Materials. All materials used in the manufacture of the cartridges shall conform strictly with the specifications referred to on the respective Naval Air Systems Command drawings unless specific approval in

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writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidders selections shall be clearly stated in the proposal.

3.3 Primary components. For the purposes of this specification, the primary components shall be:

a. primer conforming to MIL-P-20444
b. lead azide conforming to MIL-L-46225
c. lead styphnate conforming to MIL-L-757
d. HNS conforming to WS 5003

3.4 Explosive charges.

3.4.1 Quantity. All explosive charges shall be as specified on Drawing 850AS161. Charges shall be determined by weight and column heights.

3.4.2 Quality characteristics. Explosive charge weight and column height shall be a quality characteristic for statistical process control. Sampling for control charting shall be conducted by the contractor in accordance with established procedures.

3.5 Cartridge inspection.

3.5.1 Visual inspection. The complete cartridge shall be free of the visible defects noted in 4.6.1.

3.5.2 Leakage. The cartridges shall not exhibit a leak rate in excess of 10^{-5} cm^3/sec of air when tested in accordance with 4.6.2.

3.5.3 Radiographic. Cartridges shall be free of imperfections when examined in accordance with 4.6.3.

3.6 Ballistic requirements. Cartridges from each production lot, selected in accordance with 4.6.4.1, shall meet the following requirements when subjected to the test prescribed in 4.6.4.1. Any cartridge failing to meet one or more requirements shall be classified as a defective unit and the lot represented rejected.

3.6.1 Indent. Each cartridge shall be capable of causing an indentation in a dent block. When fired in the test device of 4.6.4.1, indent in the dent block shall be not less than 0.042 inches.

3.6.2 Misfire. There shall be no misfires.

3.6.3 Quality characteristics. Indent shall be a quality characteristic for statistical process control. Sampling for control charting shall be conducted by the contractor in accordance with established procedures.

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3.7 Statistical process control. Statistical process control in accordance with ANSI Z1.1, Z1.2, and Z1.3 shall be required for quality characteristics cited in this specification (explosive charge weight and column height, and indent). In addition to other requirements for this specification, only ballistic samples from lots that can demonstrate a state of statistical control shall be submitted for lot acceptance ballistic testing. Samples from lots that cannot demonstrate statistical control shall not be submitted for lot acceptance ballistic tests.

3.8 Workmanship. Cartridges shall be constructed and finished in a manner to assure compliance with requirements of this specification and associated drawings. Particular attention should be directed to dimensions, finishes, welds and seals.

4. VERIFICATION

4.1 Classification of inspections. The only inspection requirement specified herein is the conformance inspection of 4.5.

4.2 Sampling. Unless otherwise specified and when applicable (see 6.2), the sampling plans and procedures used in the determination of the acceptability of products submitted by a contractor shall be in accordance with ASQC Z1.4. Unless otherwise specified in the contract, order, or requisition (see 6.2), all samples for test purposes shall be provided at the expense of the contractor.

4.3 Lots.

4.3.1 Inspection lots. Inspection lot definition, formation, and size shall be in accordance with ASQC Z1.4.

4.3.2 Production lot. Lot numbers shall be in accordance with MIL-STD-1168. Cartridges required for all test purposes are randomly selected from the production lot and shall not be applied as a part of the quantity specified for delivery by the contract. Only primary components from a single lot shall be used in a production lot of cartridges (see 3.3). One primary component production lot may be used in more than one cartridge production lot. Cartridges selected for the ballistic test shall be delivered to the activity designated by the contracting activity (see 6.2). The designated test activity shall perform the cartridge tests and examinations listed in Table I to determine lot acceptance.

4.4 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as Critical, Major or Minor in accordance with DOD-STD-2101. Tests and examinations that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of test or examination. Tests and examinations which are not annotated with a classification code are classified as minor.

4.5 Conformance inspection. Conformance inspection shall consist of verification of the characteristics classified on the drawings listed on Drawing DL850AS161 and the tests and examinations of Table I.

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TABLE I. Conformance inspections.

Item	Nature of test	Requirement	Test method	Classification (DOD-STD-2101)
Primary component	Single lot verification	3.3	4.3.2	(M101)
Explosive charge	Type and weight verification	3.4	4.6.3.1	(M102)
Cartridge	Visual inspection	3.5.1	4.6.1	(M103)
Cartridge	Leakage	3.5.2	4.6.2	(M104)
Cartridge	Radiographic examination	3.5.3	4.6.3	(M105)
Cartridge	Ballistic requirements	3.6	4.6.4	(C1)
Cartridge	Packing, packaging, and marking	Section 5	4.5.1	(M106)

4.5.1 Packing, packaging and marking. The inspector shall ascertain that the packing and packaging of the cartridges and the container markings conform to Section 5 of this specification and as specified in the contract or order (see 6.2).

4.6 Tests.

4.6.1 Visual inspection. Each cartridge produced shall be visually examined for dented or scratched cup, incomplete or faulty weld, missing, incomplete or excessive application of sealant, incorrect or illegible markings or other visible defects which might adversely affect cartridge installation or ballistic properties. Defective cartridges shall be rejected.

4.6.2 Leakage. Each cartridge shall be leak tested in a dry gas leak tester. Cartridges which exhibit a leak rate in excess of 10^{-5} cm^3/sec of air at a pressure differential of one atmosphere shall be considered defective. Cartridges which exhibit a leak rate in excess of that specified above shall be rejected.

4.6.3 Radiographic examination. Each cartridge shall be radiographically examined in accordance with ASTM E1742 for X-ray and ASTM E545 for N-ray. The cartridge shall be positioned on its side for the most revealing exposure. All cartridges shall be identified with serial numbers prior to examination. The cartridge serial numbers shall be in consecutive order beginning with the number 001 in each production lot. The cartridges shall be arranged on tray boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the cartridges displayed thereon. The radiograph identification shall include the complete lot number, as stamped on the cartridges, and the span of serial numbers displayed. Discontinuities in serial numbers shall be identified on the radiograph. Defective cartridges found by the contractor shall be clearly identified on the radiograph and those defectives removed from the production lot. The serial numbers shall be pen or ink stamped on the cartridges with approximately 1/8-inch-high characters. The serial numbers shall be located apart from the other cartridge markings.

4.6.3.1 N-ray inspection. Each cartridge assembly shall be N-rayed to confirm the presence of the propellant, non-metal parts, no foreign hydrogenous materials, and proper assembly of all pyrotechnic/nonmetallic materials. Any observable imperfections in assembly shall be cause for rejection.

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4.6.3.2 X-ray inspection. Each cartridge shall be X-rayed to confirm the presence and proper assembly of all parts. Any observable imperfections in assembly shall be cause for rejection.

4.6.3.3 Explosive column height. The height of explosive increments within the cup shall conform to Drawing 850AS161.

4.6.4 Ballistic test.

4.6.4.1 Test. The number of cartridges selected from each production lot for the ballistic test shall be in accordance with Table I, General Inspection Level I, and Table IIA of ASQC Z1.4, plus two more cartridges for investigative purposes. The randomly selected cartridges shall be conditioned at 70 ±5°F, −65 ±5°F, or 160 ±5°F for 6 hours minimum, and fired in a test fixture conforming to Drawing 841AS287. Instrumentation shall be provided to measure pressure applied. The indent in the dent block shall be measured with a dial indicator having a probe with a 0.025 inch spherical radius tip.

4.6.4.2 Acceptance criteria. The total number of allowed defective units in a lot acceptance sample as a result of tests and inspections of 4.5 through 4.6 shall be zero. Any defects shall cause rejection of the lot represented. Screening of defects as a corrective action shall be at the contractor's expense and the method of screening shall be subject to approval by the contracting officer.

4.6.4.3 Indent. The indent shall be determined by measuring the deepest part of the indentation relative to the undisturbed surface of the dent block. The lot shall be acceptable with respect to the indent if the sample meets the requirements of 3.6.1.

4.6.4.5 Retest. There shall be no retests.

4.6.4.6 Test failure. If test failure is attributable to an assignable cause, excluding the test cartridges, original test results shall be discarded and the complete test reconducted.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activity within the Military Department or Defense Agency, or within the Military Department’s System Command. Packaging data retrieval is available from the managing Military Department’s or Defense Agency’s automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

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6.1 Intended use. The CCU-143/A impulse cartridge is designed to be installed in mechanically fired initiators and to initiate Shielded Mild Detonating Cord (SMDC).

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individualdocuments referenced (see 2.2.1, 2.2.2 and 2.3)
c. Designation of impulse cartridge (CCU-143/A)
d. Sampling plans and procedures, if other than as specified (see 4.2)
e. Activity responsible for payment of samples (see 4.2)
f. Production lot size and test activity (see 4.3.2)
g. Packaging requirements (see 5.1)
h. Items of data required for each production lot (see 6.3)
i. Shipping container marking (see 6.4)
j. That the safety precaution requirements of the "DoD Contractors' Safety Manual for Ammunition and Explosives," DOD 4145.26-M are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore," OP 5 are applicable.

6.3 Data. For the information of contractors and contracting officers, any of the data specified in subparagraphs below, applicable documents listed in section 2 of this specification or referenced lower-tier documents, need not be prepared for the Government unless specified in the contract or order (see 6.2).

6.3.1 Standard deliverables. In addition to the delivery of cartridges, the contract may require delivery of the following:

a. Radiographic plates
b. Radiographic review report
c. Radiographic sample plates and techniques
d. Requests for waiver/deviation and engineering change proposal (ECP)
e. Progress and management report
f. Test and inspection report if contractor tested
g. Ammunition data cards
h. Performance oriented packaging (POP) test report
i. Process control charts.

6.4 Marking.

6.4.1 Special marking. Marking of exterior containers should be in accordance with 49 CFR 171-179. Markings should be applied by the contractor or his agent.

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6.4.2 Normal marking. Unless otherwise specified in the contract or order (see 6.2), the marking information on unit packages and shipping containers should be as specified below:

Inner Container Marking:

a. NSN
b. NAVAIR Drawing 850AS161
c. Cartridge, Impulse, CCU-143/A
d. Quantity
e. Lot number
f. DOT/DOD Haz Class
g. UN Ser No.

Outer Container Marking:

a. NSN
b. NAVAIR Drawing 850AS161
c. Cartridge, Impulse, CCU-143/A
d. Quantity
e. Lot number
f. Placard and markings per 49 CFR 149-178

The specified marking information should be applied to the containers in accordance with the applicable provisions of MIL-STD-129. Lot number should be in accordance with MIL-STD-1168.

6.5 Subject term (keyword) listing.

Detonator
SMDC Initiator

Preparing activity:
Navy - OS
(Project 1377-0059)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-32005 (OS)	**2. DOCUMENT DATE (YYMMDD)** 971023

3. Cartridge, Impulse, CCU-143/A

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7. DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDER, INDIAN HEAD DIVISION	b. TELEPHONE *Include Area Code)* (1) Commercial (301) 743-4700 X1958 (2) AUTOVON 354-4700 X1958
c. ADDRESS *(Include Zip Code)* NAVAL SURFACE WARFARE CENTER (CODE 8410T) 101 STRAUSS AVENUE INDIAN HEAD, MD 20640-5035	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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